COMPREHENSIVE ADVERTISING SALES ANALYSIS USING MULTIPLE LINEAR REGRESSION

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Course: Mathematics for Data Science & Analytics (MDSA)

Project Type: Final-Year Capstone Dissertation

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- Supervisor:** Dr. Antigravity, Department of Data Science & Analytics

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Declaration & Certificate of Approval

Declaration

I hereby declare that this capstone project report entitled **"Comprehensive Advertising Sales Analysis using Multiple Linear Regression"** is a genuine record of my own research and analytical work conducted under the guidance of Dr. Antigravity. All mathematical formulations, dataset calculations, and visualizations are derived programmatically from the source dataset. No portion of this work has been plagiarized, and all references to academic literature have been cited.

- Date:** June 25, 2026
- Sasha** (Registration No: MDSA-2026-89)

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Certificate of Approval

This is to certify that the project report entitled **"Comprehensive Advertising Sales Analysis using Multiple Linear Regression"** is a bona fide record of work carried out by Sasha under my supervision and guidance, in partial fulfillment of the requirements for the degree of Master of Data Science & Analytics.

- Dr. Antigravity**

Project Supervisor & Professor

Department of Mathematics & Data Science

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Abstract

In the modern, data-driven business landscape, optimizing marketing expenditure across diverse advertising channels is a critical determinant of commercial success. This dissertation presents a rigorous mathematical and statistical analysis of advertising expenditure and its corresponding impact on sales revenue, utilizing a dataset representing promotional investments across three primary media: Television (TV), Radio, and Newspaper.

The analysis is structured around the five core modules of the MDSA curriculum, extended with a neural network comparative model:

1. **Module 1 (M1) - Data Preprocessing**: Inspection, duplicate/missing value handling, and outlier verification using IQR and Z-score metrics.

2. **Module 2 (M2) - Exploratory Data Analysis**: Mapping distributions, calculating descriptive moments (skewness, kurtosis), and plotting correlation matrices.

3. **Module 3 (M3) - Probability Analysis**: Modeling feature distributions, computing joint/conditional probability distributions, and Z-score target boundaries.

4. **Module 4 (M4) - Statistical Inference**: Conducting t-tests, estimating population mean confidence intervals, and evaluating variance through ANOVA.

5. **Module 5 (M5) - Regression Modeling**: Formulating the Ordinary Least Squares (OLS) Multiple Linear Regression equation and evaluating prediction metrics.

6. **Module 6 (M6) - Neural Network (Extension)**: Designing a Multi-Layer Perceptron (MLP) regressor to compare linear vs. non-linear forecast paths.

The empirical findings show that Television advertising acts as the primary driver of sales, complemented significantly by Radio, while Newspaper promotional spend shows a statistically negligible direct impact when modeled collectively. These insights provide management with a concrete mathematical framework for budget reallocation.

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1. Introduction

1.1 Background of the Topic

Marketing mix modeling (MMM) is an established analytical tool used by companies to estimate the impact of various marketing inputs on overall sales. In modern commercial structures, businesses deploy advertisements across offline channels (Television), voice channels (Radio), and traditional print media (Newspaper) to capture consumer attention. The challenge lies in quantifying the distinct and interactive contributions of each channel to optimize investment.

1.2 Importance and Significance

Understanding these mathematical relationships allows organizations to:

- Move from intuition-based marketing to evidence-based budget allocation.
- Maximize the marginal utility of promotional spend.

- Understand saturation levels (diminishing returns) of different media.

1.3 Dataset Overview

The dataset contains **200 observations** (markets) detailing advertising spend (in thousands of dollars) across three channels:

- `TV`: Promotional budget allocated to Television.
- `Radio`: Promotional budget allocated to Radio broadcast.
- `Newspaper`: Promotional budget allocated to print newspaper advertisements.
- `Sales`: Total volume of sales generated (the target output, in thousands of units).

1.4 Problem Statement

Given advertising budget details for `TV`, `Radio`, and `Newspaper`, can we construct a robust mathematical model to predict `Sales` and quantify the return on investment (ROI) of each individual channel?

1.5 Objectives

1. Clean and preprocess raw advertising data.

2. Perform Exploratory Data Analysis to establish visual and correlation trends.

3. Conduct Probability Analysis and Hypothesis Testing to confirm statistical significance.

4. Construct and validate a Multiple Linear Regression model.

5. Provide actionable mathematical budgets for corporate stakeholders.

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2. Data Cleaning and Preprocessing (M1)

2.1 Loading & Structuring

The dataset is loaded directly from `Dataset/advertising.csv` into a pandas DataFrame. The structure consists of 200 records with no missing values or duplicate records. Data types are verified as numeric float variables.

2.2 Outlier Detection

Outliers are evaluated using Z-scores and Interquartile Range (IQR) methods:

- **Z-Score Method**: Points with a Z-score greater than 3 standard deviations from the mean are flagged.

$$Z = \frac{X - \mu}{\sigma}$$

- **IQR Method**: Points outside the boundaries defined by:

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound} = Q_3 + 1.5 \times \text{IQR}$$

The IQR method flags 2 records in `Newspaper` (spending values of 89.4k and 114k), representing rare high-spend print marketing campaigns. Because these are realistic values, they are retained in the analysis.

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3. Exploratory Data Analysis (M2)

3.1 Table 2: Descriptive Statistics Summary Table

Feature	Mean	Median	Std Dev	Variance	Min	Max	Skewness	Kurtosis
TV	147.0425	149.75	85.8542	7370.95	0.70	296.40	-0.0699	-1.2265
Radio	23.2640	22.90	14.8468	220.43	0.00	49.60	0.0942	-1.2604
Newspaper	30.5540	25.75	21.7786	474.31	0.30	114.00	0.8947	0.2395
Sales	15.1305	16.00	5.2839	27.919	1.60	27.00	-0.0737	-0.6554

3.2 Bivariate Linear Correlation Metrics

Feature	TV	Radio	Newspaper	Sales
TV	1.0000	0.0548	0.0566	0.9012
Radio	0.0548	1.0000	0.3541	0.3496
Newspaper	0.0566	0.3541	1.0000	0.1580
Sales	0.9012	0.3496	0.1580	1.0000

3.3 Bivariate Bounding Interpretation

- **TV vs. Sales**: Strong linear association ($r = 0.9012$). Budget increases translate directly to predictable sales volume growth.
- **Radio vs. Sales**: Moderate correlation ($r = 0.3496$). While linear, it shows higher variance than TV.
- **Newspaper vs. Sales**: Weak correlation ($r = 0.1580$). The distribution is highly skewed, and higher spend does not reliably increase sales.

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4. Probability Analysis (M3)

4.1 Normal Fit and Marginal Estimates

Fitting a normal distribution curve to target variable Sales yields $\mu = 15.1305$ and $\sigma = 5.2839$.

Using a target sales success threshold of 15.0k units:

- **Theoretical Success Probability**: $P(\text{Sales} > 15.0\text{k}) = 0.4485$ (44.85% based on Normal fit).

- **Conditional Probability TV Spend**: $P(\text{Sales} > 15.0\text{k} \mid \text{TV} > \text{median}) = 0.8200$ (82% probability of success when TV budget is above median).
- **Conditional Probability Newspaper Spend**: $P(\text{Sales} > 15.0\text{k} \mid \text{Newspaper} > \text{median}) = 0.4900$ (49% probability, showing negligible difference from baseline random chance).

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5. Statistical Inference (M4)

5.1 Two-Sample t-Test

To evaluate the impact of TV spend on Sales, we divide markets by median TV spend and test:

- H_0: $\mu_{\text{high\_tv\_sales}} = \mu_{\text{low\_tv\_sales}}$
- H_1: $\mu_{\text{high\_tv\_sales}} > \mu_{\text{low\_tv\_sales}}$

The t-test yields $t = 15.545$ and $p = 5.87 \times 10^{-37}$. We reject the null hypothesis (H_0), confirming that higher TV spend generates significantly higher mean sales.

5.2 Analysis of Variance (ANOVA)

Categorizing TV budget into three groups (Low, Medium, High) and running ANOVA yields $F = 204.6$ and $p = 1.04 \times 10^{-48}$. The mean sales across these three budget categories differ significantly and non-randomly.

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6. Multiple Linear Regression Model (M5)

6.1 OLS Equation and Coefficients

Using an 80/20 train-test split, we fit the OLS Multiple Linear Regression model:

$$\text{Sales} = 4.7141 + 0.0545 \times \text{TV} + 0.1009 \times \text{Radio} + 0.0043 \times \text{Newspaper}$$

Table 4: Dynamic Regression Coefficients & Direction

Feature	Coefficient (β)	Direction	Business Interpretation
TV	0.0545	Positive	Each $1k increase in TV spend predicts a sales increase of 54.5 units, holding other channels constant.
Radio	0.1009	Positive	Each $1k increase in Radio spend predicts a sales increase of 100.9 units, holding other channels constant.
Newspaper	0.0043	Positive	Each $1k increase in Newspaper spend predicts a sales increase of 4.3 units, showing negligible statistical return.

Table 5: Model Performance Dashboard DataFrame

Metric	Formula	Score Value	Interpretation
MAE	$\frac{1}{n} \sum \vert y_i - \hat{y}_i \vert$	1.4608	On average, predictions deviate from actual sales by 1.46 thousand units.
MSE	$\frac{1}{n} \sum (y_i - \hat{y}_i)^2$	2.9013	The average squared error is 2.90.
RMSE	$\sqrt{\text{MSE}}$	1.7033	The standard deviation of model residuals is 1.70 thousand units.
R2 Score	$1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$	0.9059	The model explains 90.59% of the variance in sales.

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7. Module 6 - Artificial Neural Network (Extension)

7.1 Model Architecture & Implementation

To test for non-linear relationships, we implement a Feedforward Multi-Layer Perceptron (MLP) regressor:

- **Input layer**: 3 normalized features (`TV`, `Radio`, `Newspaper`).
- **Hidden layer 1**: 8 neurons with ReLU activation.
- **Hidden layer 2**: 4 neurons with ReLU activation.
- **Output layer**: 1 neuron (Linear activation).
- **Optimizer**: Adam optimizer.

Table 8: Model Performance & Architectural Comparison

Metric	Multiple Linear Regression	Artificial Neural Network	Difference (ANN - MLR)	Better Model
MAE	1.4608	0.7712	-0.6896	**ANN** (Lower error)
MSE	2.9013	1.0423	-1.8590	**ANN** (Lower error)
RMSE	1.7033	1.0209	-0.6824	**ANN** (Lower error)
R2 Score	0.9059	0.9662	+0.0603	**ANN** (Higher variance explained)
Training Time	< 0.001s	0.184s	+0.183s	**MLR** (Faster)
Model Complexity	Low	Medium	-	**MLR** (Simpler)
Interpretability	High (Direct Coefficients)	Low (Blackbox weights)	-	**MLR** (Explainable)
Business Suitability	High for budget design	High for tactical forecasting	-	Context Dependent

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8. Managerial Recommendations

1. **Reallocate Newspaper Budgets**: Redraw the print marketing budget. Newspaper spend shows a statistically negligible impact on sales ($\beta_3 = 0.0043$), so reallocate up to 80% of these funds to more active channels.

2. **Prioritize Radio for High-Efficiency ROI**: Radio advertising yields the highest marginal return per dollar ($\beta_2 = 0.1009$). Maximize Radio campaigns in key local markets.

3. **Establish TV Spend as the Scale Foundation**: Maintain TV advertising as the primary driver of scale. TV spend is highly correlated with sales ($r = 0.9012$) and forms the backbone of long-term brand equity.

4. **Enforce Minimum TV Spends**: Markets with above-median TV spend have an 82% probability of achieving high sales ($>15\text{k}$). Enforce minimum TV spend thresholds to reduce sales variance.

5. **Deploy Hybrid Modeling**: Use Multiple Linear Regression to design marketing mix budgets and explain ROI to stakeholders. Use the Neural Network model for high-precision regional sales forecasting.

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9. References

- Farris, P. W., Bendle, N. T., Pfeifer, P. E., & Reibstein, D. J. (2010). *Marketing Metrics: The Definitive Guide to Measuring Marketing Performance*. Pearson Prentice Hall.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning: with Applications in R*. Springer.
- Montgomery, D. C., Peck, E. A., & Vining, G. G. (2012). *Introduction to Linear Regression Analysis* (5th ed.). John Wiley & Sons.
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
- Kaggle Advertising Dataset. Retrieved from https://www.kaggle.com/datasets/ashydv/advertising-dataset